

Session 14 Quiz - AI That Reads - Natural Language Processing (NLP)

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57351]

To process language, AI first splits sentences into small pieces called _____.

- A. alphabets
- B. bytes
- C. tokens
- D. signals

Correct Answer: C. tokens

Explanation: Tokens are the individual 'bricks' of language, such as words or punctuation marks, that AI processes one by one. This is like breaking a sentence into LEGO pieces. Alphabets are letters, bytes are computer storage units, and signals are for communication systems.

Q2 [Type: MCQ | Difficulty: Easy | QID:57352]

Identifying that 'Tokyo' is a place and 'Sara' is a person is known as Named Entity _____.

- A. Generation
- B. Recognition
- C. Translation
- D. Correction

Correct Answer: B. Recognition

Explanation: Named Entity Recognition (NER) involves the AI identifying and categorizing specific nouns into groups like people, locations, dates, or organizations. Generation creates new text, translation converts languages, and correction fixes errors.

Q3 [Type: MCQ | Difficulty: Easy-Medium | QID:57353]

The step where AI figures out the 'why' or the feeling behind words is called _____.

- A. Understanding
- B. Scanning
- C. Printing
- D. Breaking

Correct Answer: A. Understanding

Explanation: Also known as Natural Language Understanding (NLU), this step is where the AI attempts to determine intent, sentiment, and connections between ideas. Scanning is looking for keywords, printing is output, and breaking is tokenization.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57354]

When AI writes a response back to you one piece at a time, it is performing Natural Language _____.

- A. Grammar
- B. Grouping
- C. Geography
- D. Generation

Correct Answer: D. Generation

Explanation: Natural Language Generation (NLG) is the 'writing back' phase of the NLP recipe where the AI builds a response token by token. Grouping, geography, and grammar are not the correct terms for this output process.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:57355]

If a sentence like 'I saw her duck' can mean two different things, it is an example of _____.

- A. Alphabetizing
- B. Ambiguity
- C. Accuracy
- D. Alliteration

Correct Answer: B. Ambiguity

Explanation: Ambiguity occurs when language is unclear or has multiple meanings. 'Duck' could mean the bird or the action of lowering the head. This is a major challenge for AI because it has to figure out which meaning is intended.

Q6 [Type: True/False | Difficulty: Easy | QID:57356]

True or False: AI reads whole sentences all at once just like humans do.

- A. False
- B. True

Correct Answer: A. False

Explanation: AI must break sentences into smaller units called tokens before it can begin processing them. It does not read whole sentences at once like humans do.

Q7 [Type: True/False | Difficulty: Easy-Medium | QID:57357]

True or False: Sarcasm is easy for computers to understand because they only look at the literal meaning of words.

A. False

B. True

Correct Answer: A. False

Explanation: Sarcasm is actually very hard for AI because the intended meaning is the opposite of the literal word choice. For example, "Oh great" might actually mean something is bad. AI struggles with this.

Q8 [Type: True/False | Difficulty: Easy | QID:57358]

True or False: Sentiment analysis is the NLP job used to figure out if a review is happy or angry.

A. False

B. True

Correct Answer: B. True

Explanation: Sentiment analysis focuses on detecting the emotional tone or feeling behind a piece of text, such as positive, negative, or neutral. It is commonly used for product reviews or social media posts.

Q9 [Type: True/False | Difficulty: Easy | QID:57359]

True or False: Google Translate uses the same 'Break → Understand → Generate' recipe as ChatGPT.

A. False

B. True

Correct Answer: B. True

Explanation: Almost all NLP tools follow this core framework (tokenization, understanding, generation) to process and output human language, including translation tools and chatbots.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:57360]

True or False: AI labels tokens as 'Nouns' or 'Verbs' to help it figure out what the sentence means.

A. False

B. True

Correct Answer: B. True

Explanation: Labeling word types (parts of speech) allows the AI to understand the structure and action within a sentence, which helps with meaning. For example, knowing which word is the verb tells the AI what action is happening.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57361]

Why does the source material compare AI tokens to LEGO bricks?

- A. To explain that AI is a game for children.
- B. Because tokens are made of colorful plastic.
- C. To show how AI builds a sentence piece by piece.
- D. Because you can buy tokens at a toy store.

Correct Answer: C. To show how AI builds a sentence piece by piece.

Explanation: Tokens are the individual components that AI uses to construct or deconstruct language, much like building with LEGO bricks. The analogy highlights how small pieces come together to make a whole.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:57362]

Which NLP job is being used when Gmail flags an email as 'Spam'?

- A. Paraphrasing
- B. Translation
- C. Speech-to-Text
- D. Sentiment / Classifying

Correct Answer: D. Sentiment / Classifying

Explanation: The AI analyzes the content to categorize it based on patterns typically found in unwanted messages. This is classification (and sometimes sentiment analysis), deciding what "kind" of message it is.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57363]

How many tokens would the AI see in the sentence: 'The cat sleeps.'?

- A. 1
- B. 4
- C. 3
- D. 12

Correct Answer: B. 4

Explanation: The AI counts 'The', 'cat', 'sleeps', and the period '.' as individual tokens. Punctuation marks are usually treated as separate tokens because they carry meaning (end of sentence).

Q14 [Type: MCQ | Difficulty: Easy-Medium | QID:57364]

What is the specific goal of the 'NLU' part of the NLP recipe?

- A. To translate words into a different font.
- B. To physically print the words on a screen.
- C. To figure out the meaning and intent behind the words.
- D. To count how many letters are in each word.

Correct Answer: C. To figure out the meaning and intent behind the words.

Explanation: Natural Language Understanding (NLU) focuses on the logic, feelings, and connections within the text. It goes beyond just recognizing words to understanding what they mean in context.

Q15 [Type: MCQ | Difficulty: Easy | QID:57365]

Which of these tools is specifically designed to help you summarize long articles?

- A. QuillBot
- B. YouTube
- C. Google Translate
- D. NotebookLM

Correct Answer: D. NotebookLM

Explanation: NotebookLM is a tool used to organize sources and can generate short summaries of long documents. QuillBot paraphrases, Google Translate translates, and YouTube is a video platform.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:57366]

Why is the idiom 'It's raining cats and dogs' difficult for AI?

- A. Because the AI thinks it's a math problem.
- B. Because it uses figurative language that isn't literal.
- C. Because the sentence is too short for the AI to read.
- D. Because the AI doesn't know what a cat is.

Correct Answer: B. Because it uses figurative language that isn't literal.

Explanation: AI often struggles with phrases where the words don't mean what they say literally. Idioms require understanding cultural context and non-literal meaning, which is hard for AI.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57367]

In the sentence 'Ugh, my team lost AGAIN today,' what sentiment would the AI likely detect?

- A. Positive
- B. Sarcastic
- C. Neutral
- D. Negative

Correct Answer: D. Negative

Explanation: The words 'Ugh' and 'lost' are strong indicators of a sad or frustrated emotional tone. Sentiment analysis would classify this as negative, not positive, sarcastic, or neutral.

Q18 [Type: MCQ | Difficulty: Easy | QID:57368]

What does QuillBot do when it 'Paraphrases' a sentence?

- A. It keeps the meaning but changes the words.
- B. It turns the text into a video.
- C. It deletes all the nouns in the sentence.
- D. It translates the sentence into Spanish.

Correct Answer: A. It keeps the meaning but changes the words.

Explanation: Paraphrasing involves rewriting a statement in a different way while maintaining the original intent. It does not delete nouns, turn text into video, or translate to another language.

Q19 [Type: MCQ | Difficulty: Easy-Medium | QID:57369]

Which challenge for AI involves the word 'football' meaning something different in the US vs. the UK?

- A. Sarcasm
- B. Idioms
- C. Different Cultures
- D. Spelling

Correct Answer: C. Different Cultures

Explanation: Regional differences in language usage (cultural context) can confuse AI that hasn't been trained on those specific contexts. In the US, football means American football; in the UK, it means soccer.

Q20 [Type: MCQ | Difficulty: Medium | QID:57370]

What is the main difference between NLU and NLG?

- A. NLU is fast, and NLG is slow.
- B. NLU is about listening/reading, while NLG is about talking/writing.
- C. NLU uses words, and NLG uses pictures.
- D. NLU is for math, and NLG is for English.

Correct Answer: B. NLU is about listening/reading, while NLG is about talking/writing.

Explanation: Natural Language Understanding (NLU) is the input and analysis phase (reading/listening), while Natural Language Generation (NLG) is the output and creation phase (writing/talking). They are two halves of the NLP process.